

Seminar Quiz 6

The Economy

Sciences Po Paris - Reims

15 minutes. No extra papers. You are free to use calculators.

Exercise 1

Consider two countries, Liptonia and Borelia. Both countries produce only two goods: boron and tea. The following table shows the number of hours needed to produce one unit of each good in each country.

	Liptonia	Borelia
boron	12 hours	3 hours
tea	6 hours	5 hours

1. Show if any of these countries have an absolute advantage. Then, draw a matrix as above and write the relative prices for each country. Which country has the comparative advantage in which good? (2 points)

- Borelia has absolute advantage in both goods because it takes a shorter amount of time to produce both.

	Liptonia	Borelia
boron	$\frac{12}{6}$	$\frac{3}{5}$
tea	$\frac{6}{12}$	$\frac{5}{3}$

- Looking at the relative prices, Liptonia has comparative advantage in tea whereas Borelia has comparative advantage in boron.

2. Assuming Liptonia has 480 hours and Borelia has 150 hours at their disposal, draw a graph showing the feasible consumption frontiers for both countries in an autarky (with boron on the X axis). Then, add another consumption frontier where there is trade and the world relative price of boron is 1. (3 points)

Calculate corner points for the feasible consumption frontiers for both countries:

Liptonia: 40 units of boron or 80 units of tea

Borelia: 50 units of boron or 30 units of tea

As trade happens, the world relative price converges to 1. Liptonia specializes in tea and Borelia specializes in boron, creating a higher feasible consumption frontier for both countries.

Exercise 2

Recall the Phillips curve equation:

$$\pi_t = \pi_{t+1}^e - \beta(u_t + \bar{u}) + c_t$$

1. Assume that there is a significant development in AI technology, greatly increasing productivity across multiple sectors. Explain the immediate effects of this development on inflation, using the equation. (2 points)

An increase in productivity can either increase unemployment u_t or decrease costs c_t . As a result of either, inflation will decrease according to the Phillips curve equation.

2. Capital-owners want to implement this technology as soon as possible, but they need money to do so. The central bank in this economy also wants to incentivize the adoption of this technology. What type of monetary policy would the central bank follow? Draw a supply and demand graph for the loan market to explain your reasoning. Show the change on both sides. (3 points)

Investing in new technology will increase demand for loans and the central bank will incentivize by following an expansionary monetary policy, increasing the supply for loans. This will push both curves to the right.

